Parallel Ranger Deck in Isaac Lab

Cyber Physical Systems Programming – Project

Students: Luca Samorì • Alexandru Zaporojanu • Tommaso Tieri

Teacher: Prof. Andrea Acquaviva | **Supervisor:** Dott. Sebastiano Mengozzi

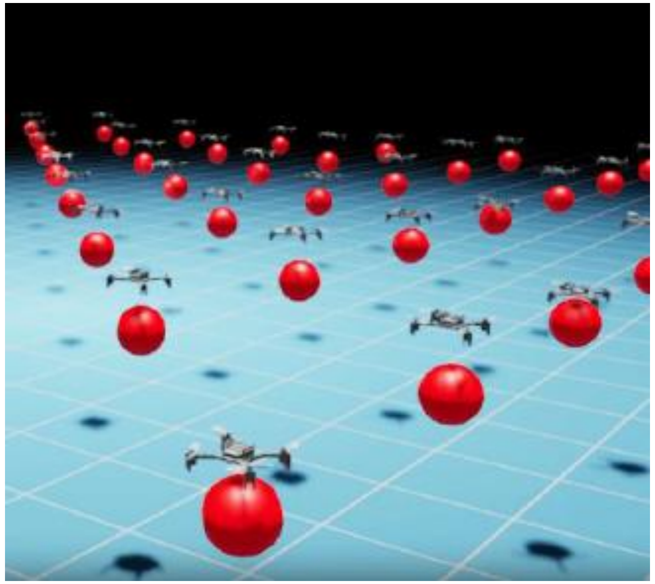
The Project

- **Goal**

  Extend the **Isaac Lab sensors library** with a custom ranger deck sensor tailored for navigation tasks.

- **Description**

  Create a custom software library to implement the Multi-Ranger Deck sensor within the Isaac Lab simulation environment, ensuring accurate environmental perception for drones.

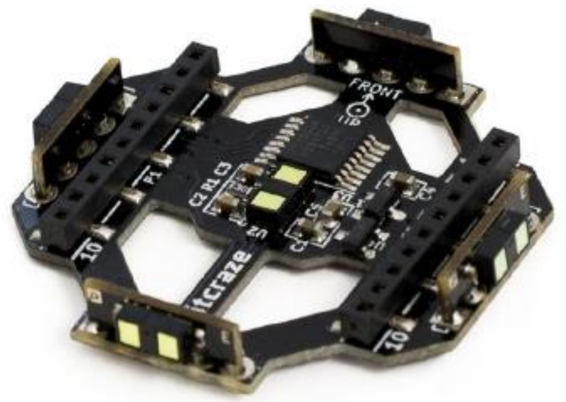


bitcraze

NVIDIA

How it works: Multi-Ranger Deck

> *"The Crazyflie Multi-ranger deck gives the Crazyflie 2.X the ability to detect objects around it. This is done by measuring the distance to objects in the following **5 directions: front/back/left/right/up** with mm precision up to 4 meters."* [1]



Data Sheet Sensor:

5 Axes

Omnidirectional sensing

4 Meters

Maximum range

mm Precision

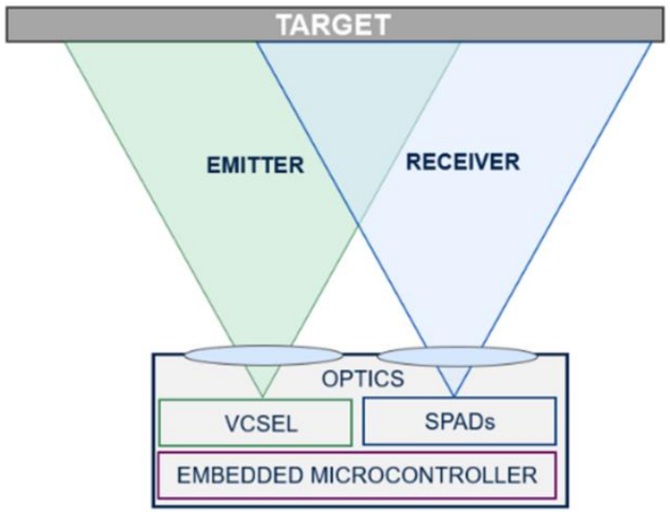
High accuracy

VL53L1x

5x ToF Sensors

Sensor **Specifications**

- **Mechanical:** Weight: 2.3g | Size (WxHxD): 35x35x5mm.
- **Emitter:** 940 nm invisible laser emitter (Class 1).
- **Receiver:** SPAD (single photon avalanche diode) receiving array with integrated lens.
- **Field of View:** 27° typical full field of view (FoV).
- **ROI:** Programmable region of interest allows reduction of the sensor FoV. [2][3]



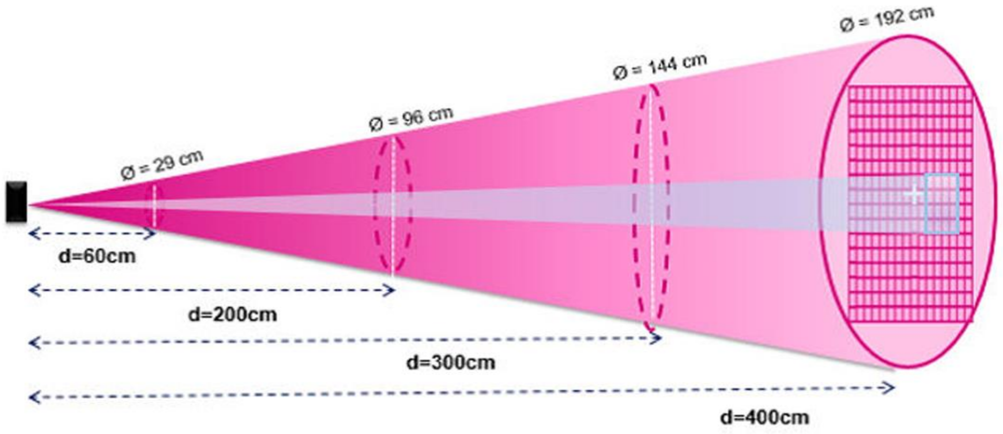
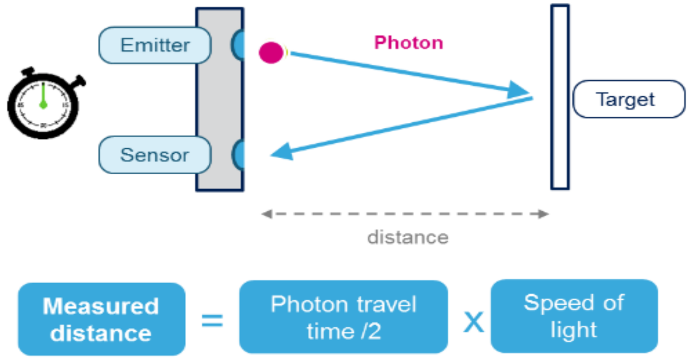


Figure: 27° field of view (FoV) down to 15 [3]

Time of Flight – FlightSenseTM Principle

Basic Principle



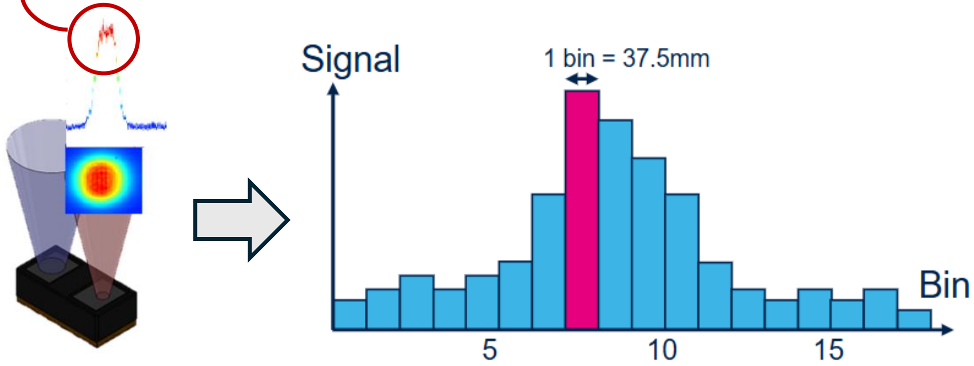
1cm round-trip at 67ps!

97% of the optical power

in a cone of

27°.

In Practice



Basic Principle

It measures the time it takes for a photon to bounce back.

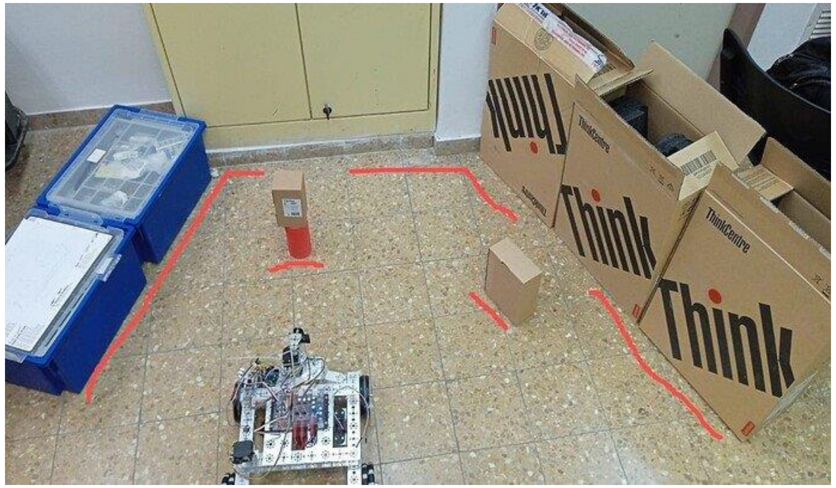
If the cone hits multiple objects at different distances, the histogram will show multiple peaks.

The final measured distance is an average derived from these significant peaks [4]

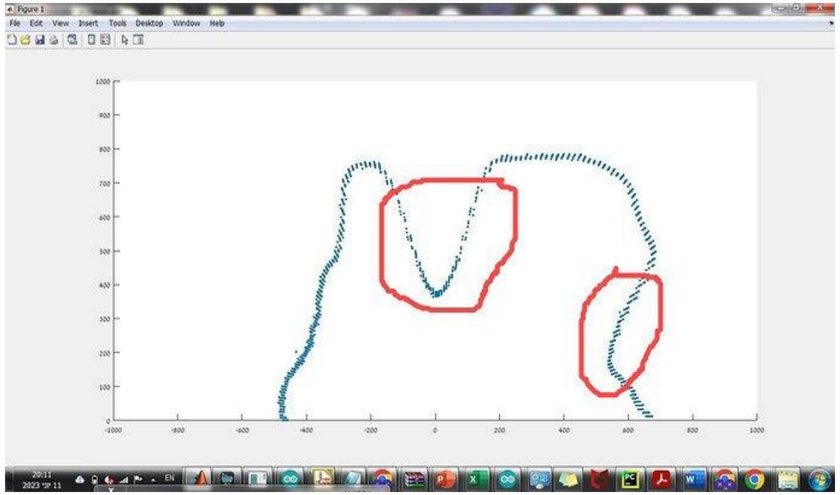
Physical Behavior vs Simulation Gap

- One ray do not simulate the actual behavior of the sensor
- We need to fill the Real-to-Sim Gap

Practical example: Generation of a map [5]



Expected map would approximate to red lines

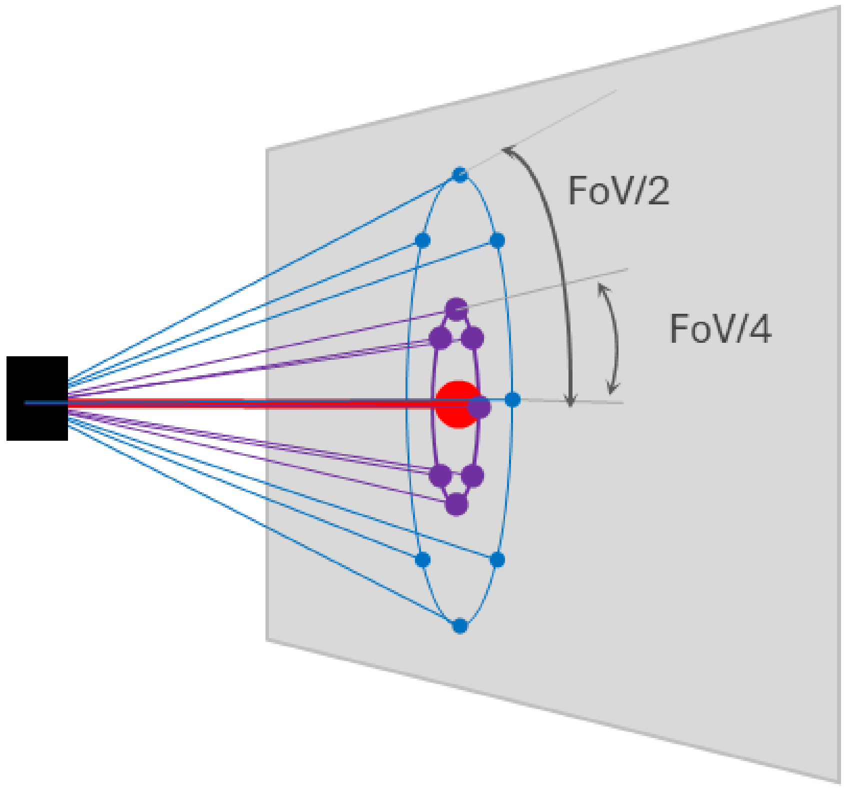


In practice, we get an average when multiple objects are present in the FoV

Physical Behavior vs Simulation Gap

The Real 2 Sim Gap:

- Rays distributed on the surface of **2 Concentric cones** and **1 central ray** receive distance measures.
- The final output is a **weighted average** with higher weights given to central beams.
- Rays number can be variated.
- FoV can take whatever value, and cones will adapt (27°-15° is physical limit).
- This physical choices are important for the code

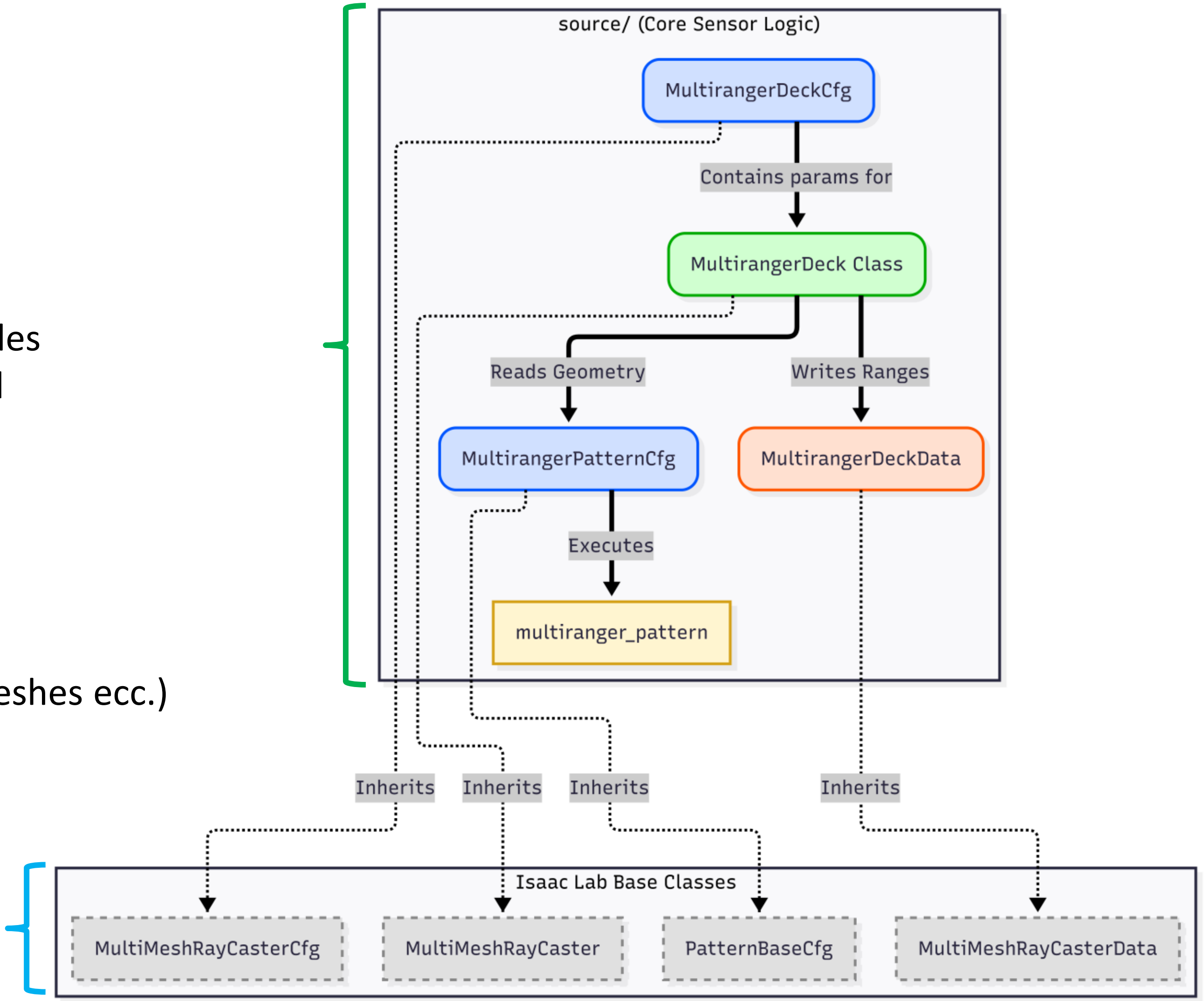


Code Structure

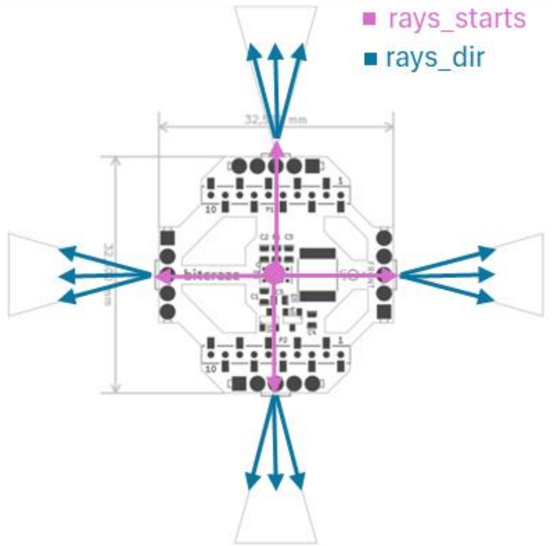
- Our code extends the base class; it's the «brain» that customizes
the sensor's behavior

- Isaac Lab, using the MultiMeshRayCaster class, provides the infrastructure (ability to interact with meshes and compute everything in parallel on the GPU).

- <u>Benefits:</u>
  - Dynamic mesh detection
  - Multimesh detection (cones, sphere, costume meshes ecc.)
  - Ease of usage
  - GPU-Accelerated Parallelization.



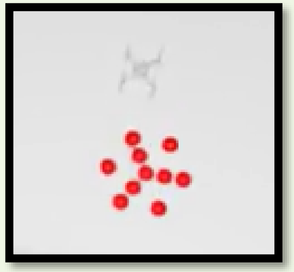
How the 5 Cones Were Created



- ray_starts coordinate of the ray origin.
- ray_dirs unit vector of the direction.

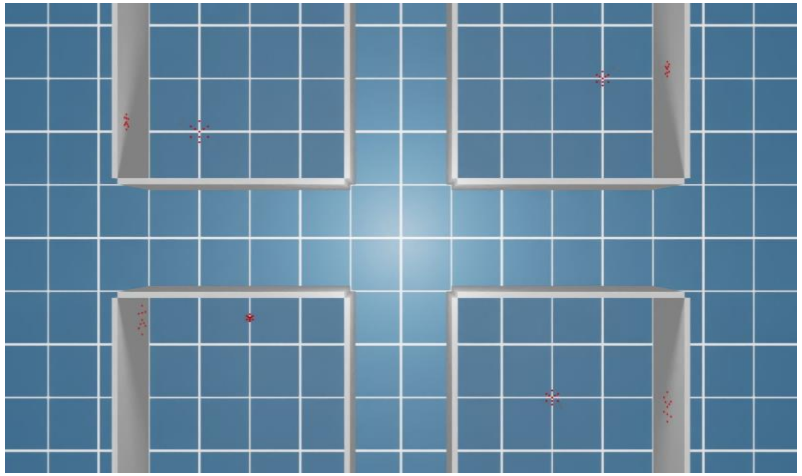
Defines the geometry for every sensor: 10 ray for each sensor (50 rays)

The algorithm automatically splits in 1 central ray, 1/3 inner rays and 2/3 outer rays.



- Gets the 50 impact points from the GPU
- Transforms the 50 points into 5 distances using a **weighted average**
- Returns a **Tensor** (N, 5). A table where each row is a drone and the 5 columns are the distances in the 5 directions

Demo 1: Parallelization & Validation



- **Accuracy** of measurements
- 4 drone spawn with their respective sensors
- Plotted the different measurements of all the sensors

Isaac Lab Parallelism

Test showing the operation of both the sensor measurement and the parallelization, a core feature of the Isaac Lab simulation environment.

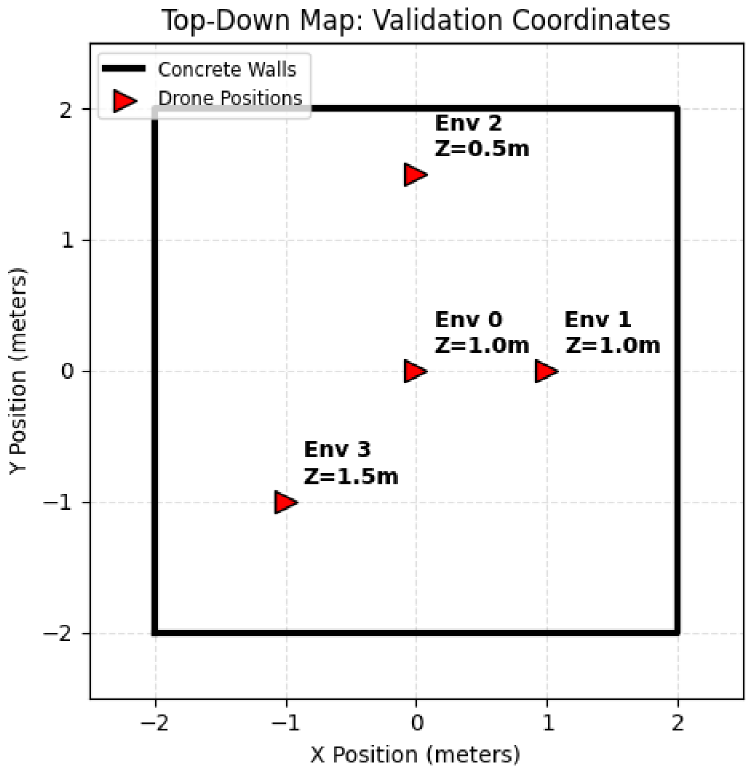
Expected vs Measured

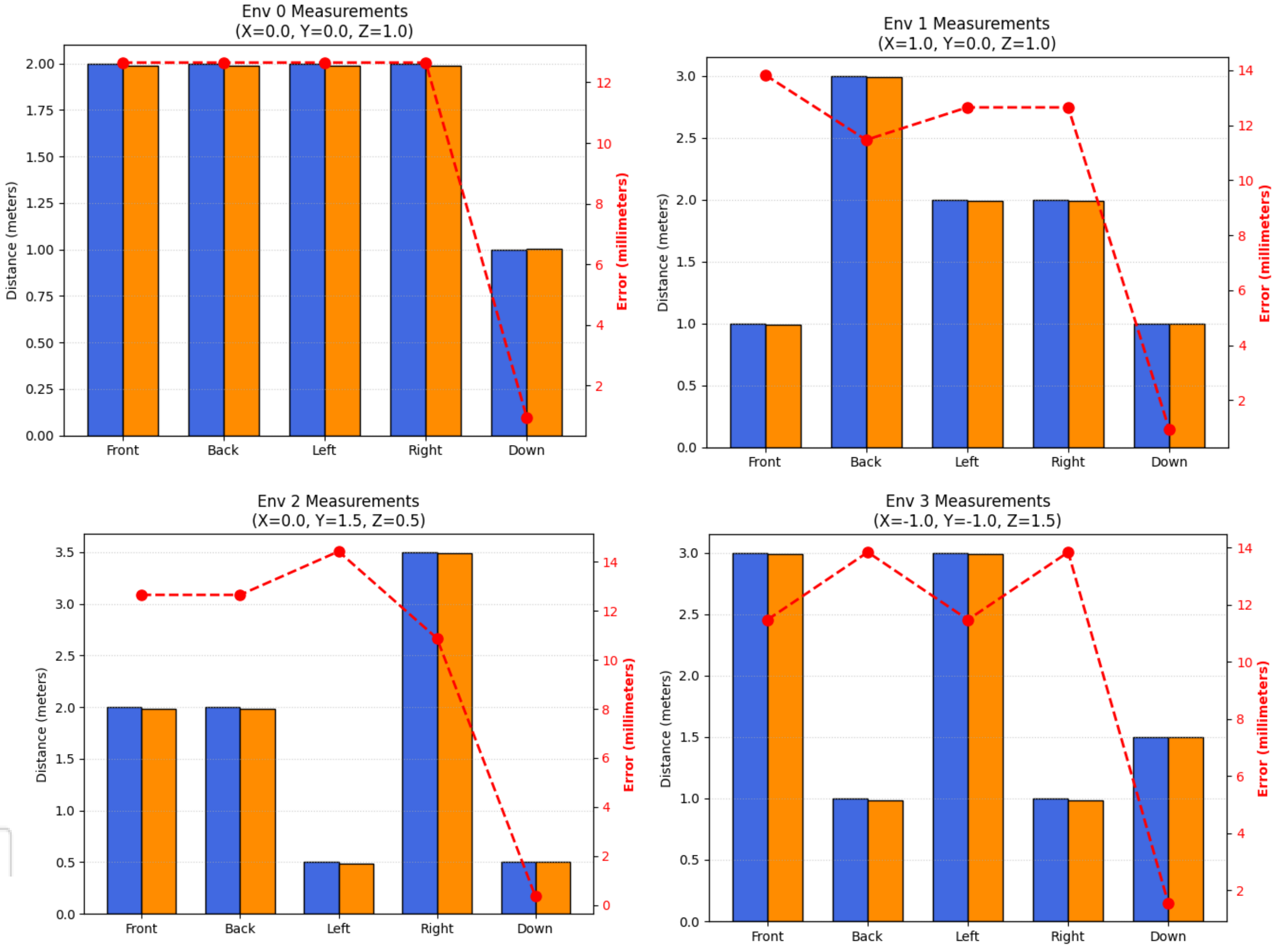
Comparison between theoretical distance (concrete walls) and the simulated ToF measurement.

Error Tracking

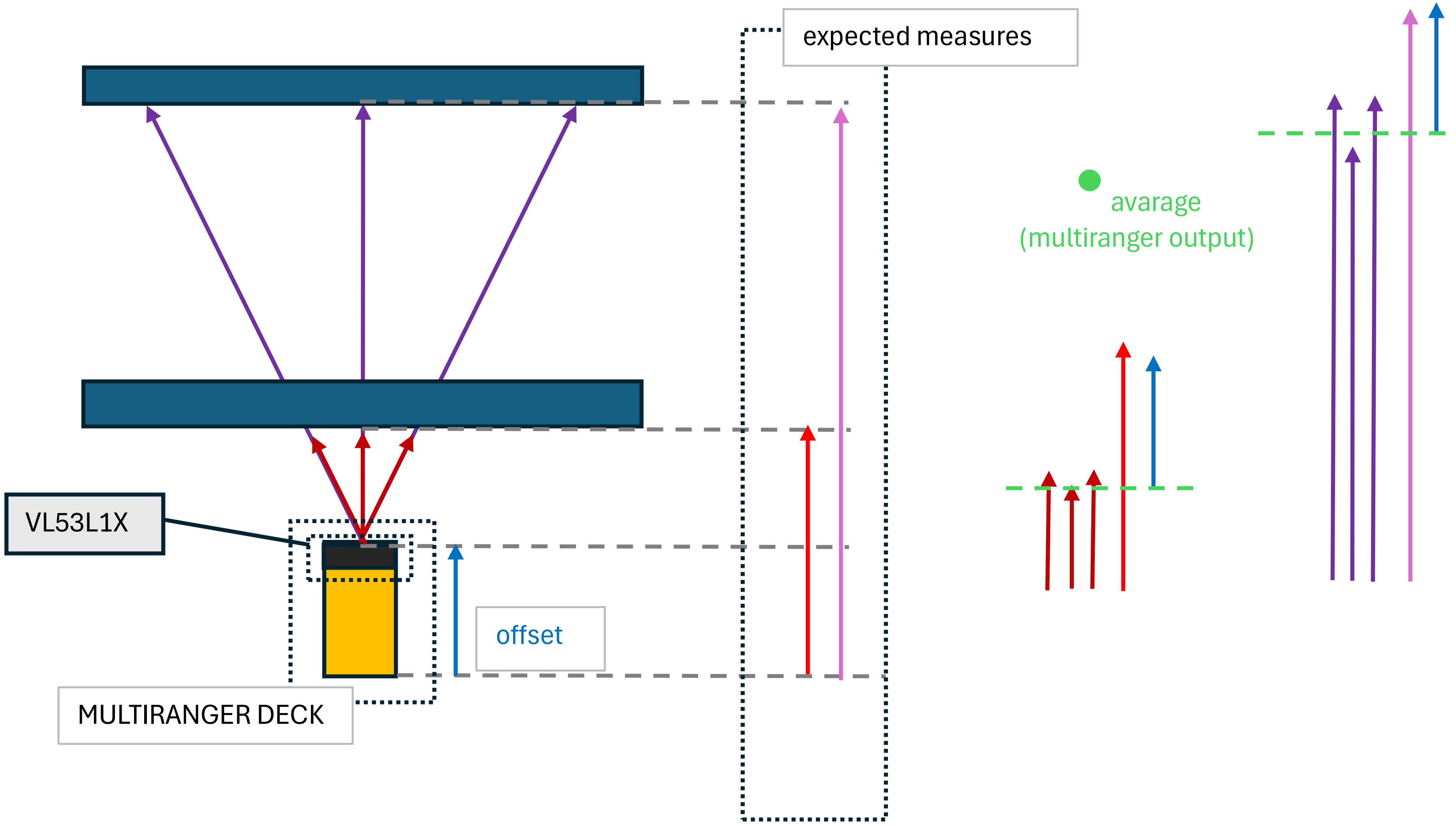
Error in millimeters plotted across all 5 directions (Front, Back, Left, Right, Down).

Demo 1: Results





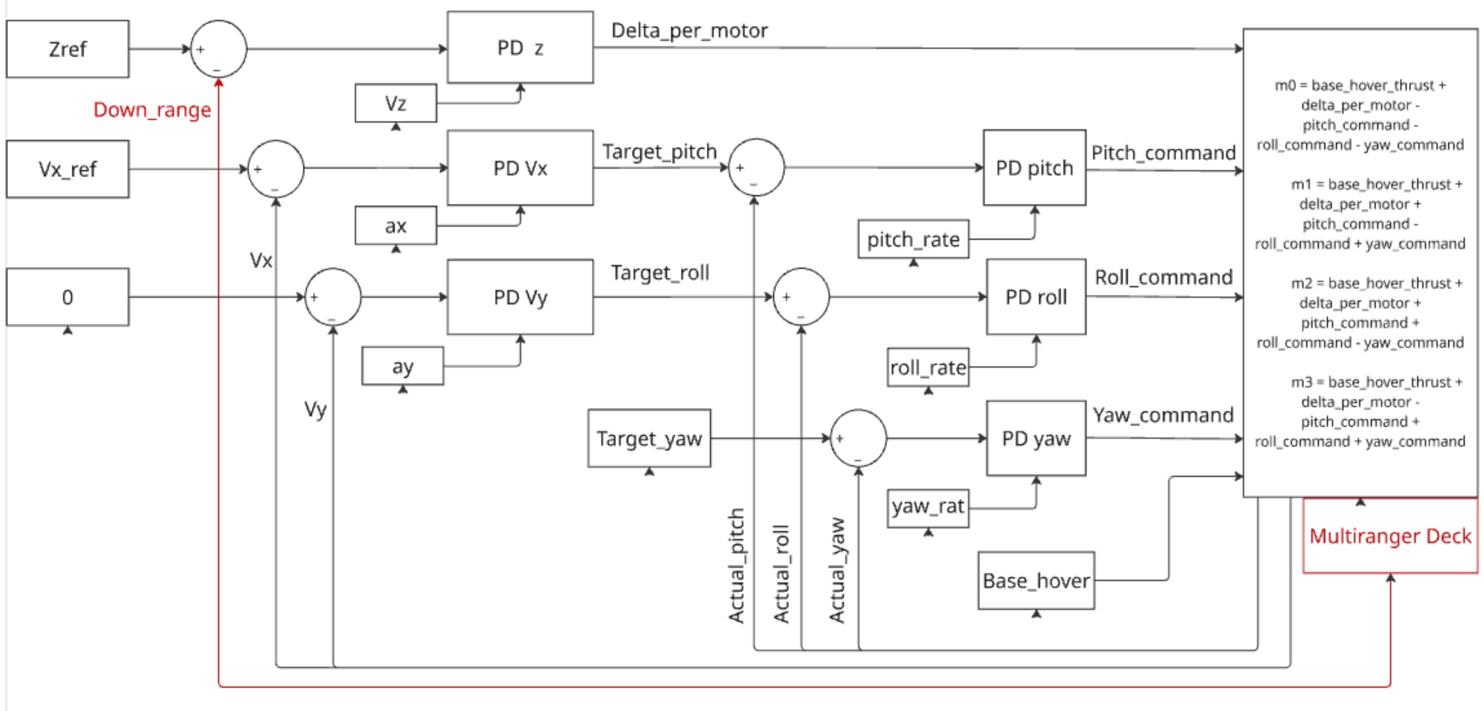
Demo 1: Results



Demo 2: Flight Controller

Stairs Environment

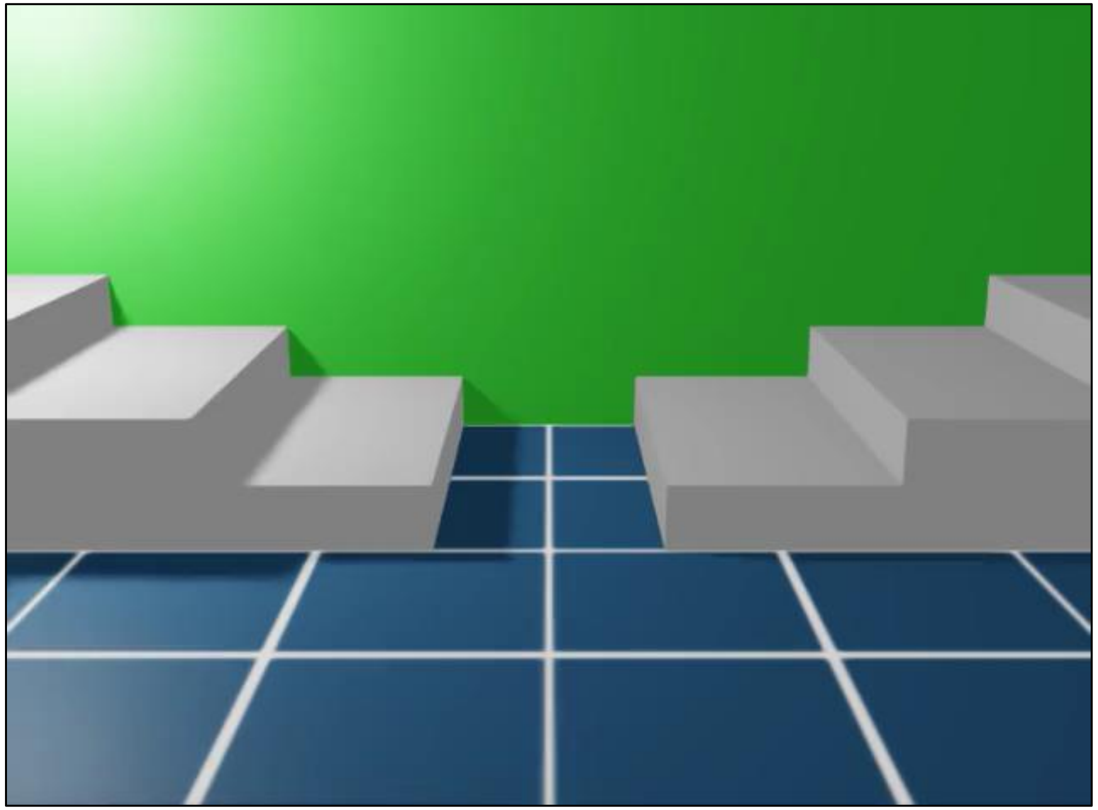
- The drone navigates over a set of stairs. We implemented a control loop to maintain a stable height.

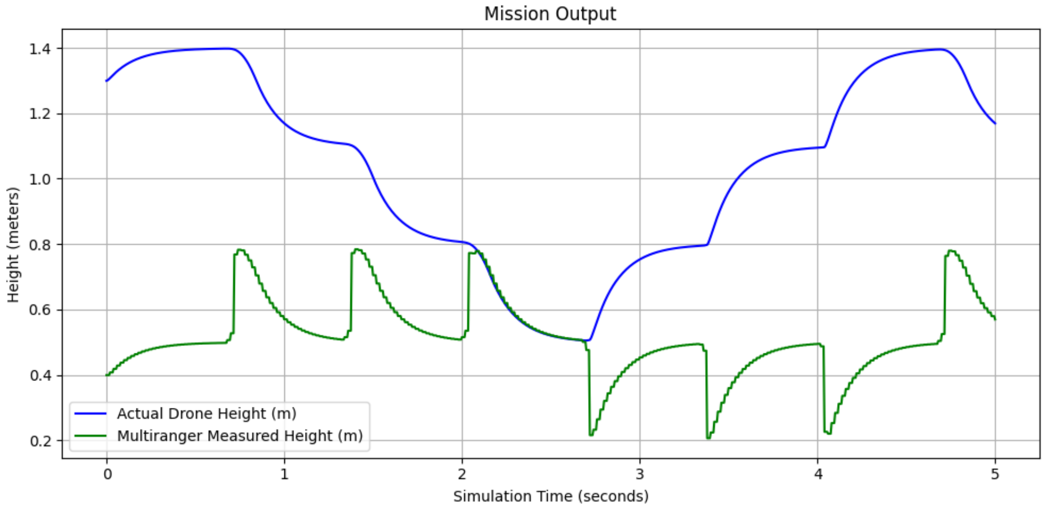


Control Scheme Loop

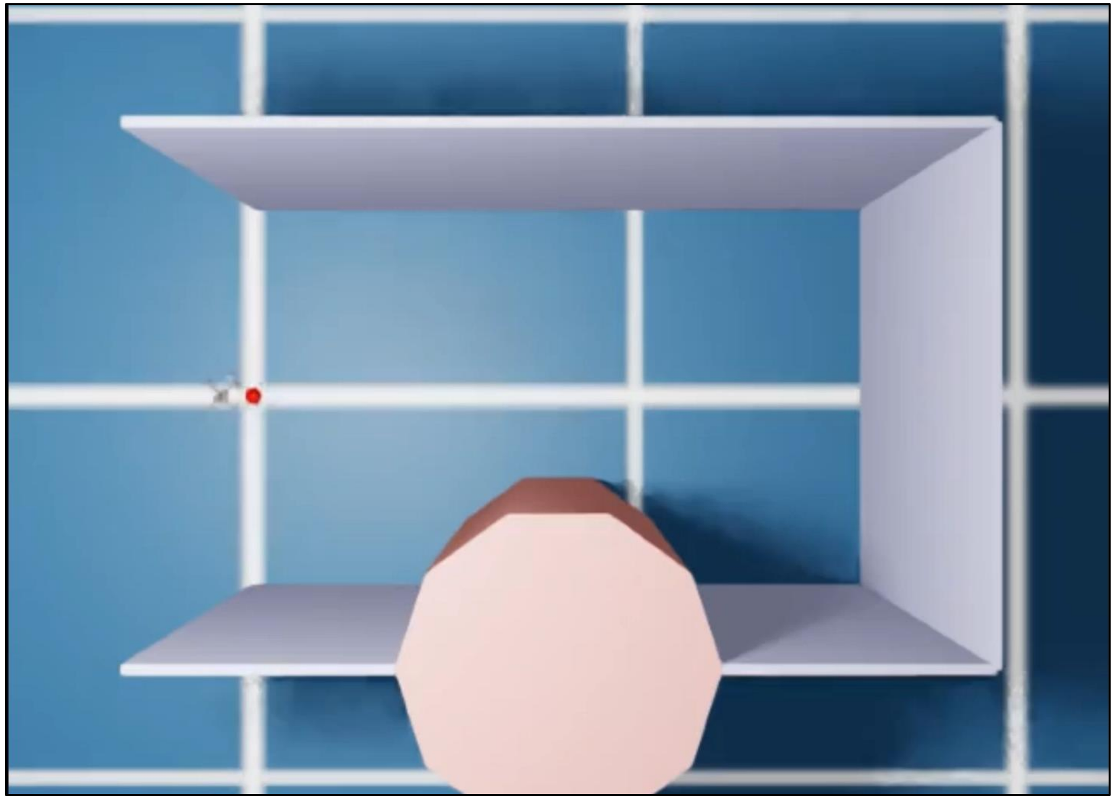
- **Setpoint:** Desired height (e.g., 0.5m)
- **Sensor (Multi-ranger Down):** Reads actual altitude relative to the stairs.
- **Error:** Setpoint - Sensor Reading.
- **PID Controller:** Adjusts vertical thrust to minimize error.

Demo 2: Results





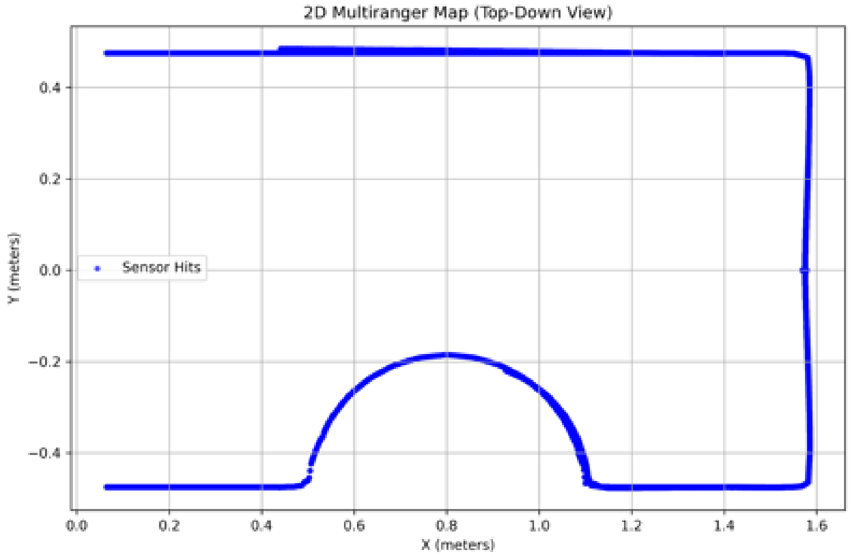
Demo 3: 2D Mapping



Mapping the Environment

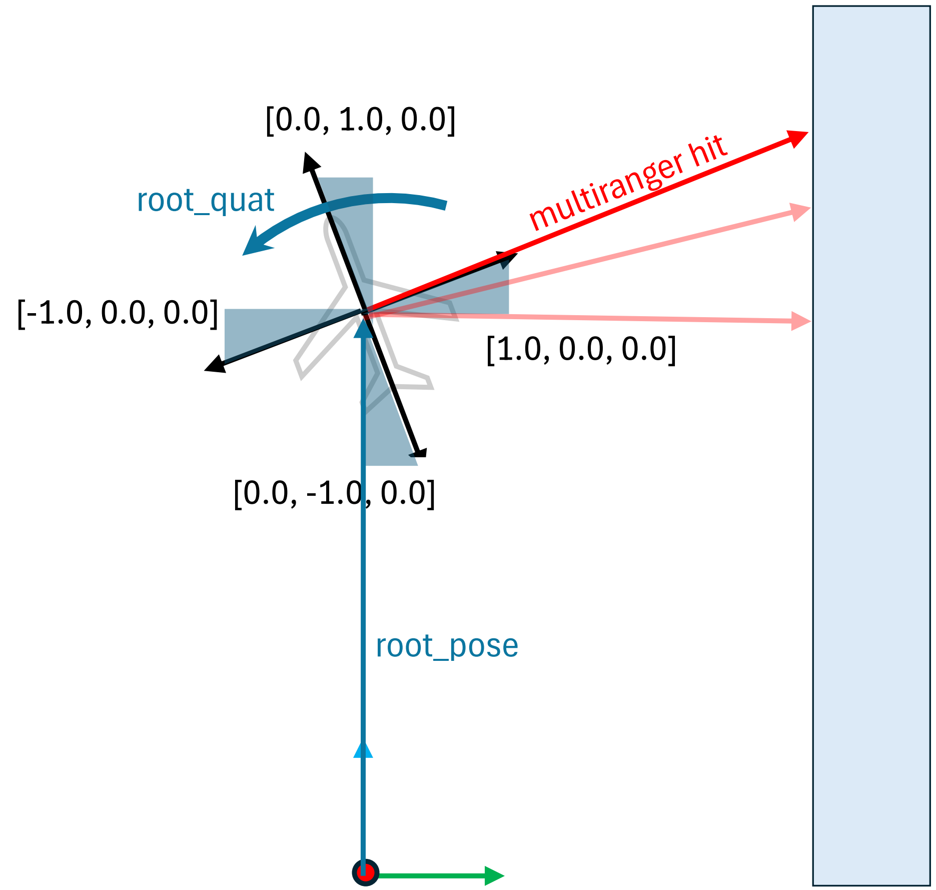
As the drone moves through a corridor with rounded and sharp obstacles, the side lasers continuously gather point data.

- **Sensor Hits:** Plotted as a point cloud on an X-Y plane.
- Demonstrates the geometric fidelity of the sensor wrapping over complex shapes (like the semi-circular wall obstacle).



Demo 3: 2D Mapping

$$point\ claude = root\ pose + R(base\ unit\ vectors) \cdot multiranger\ hit$$



Conclusions and Future Works

- Add the functionality of being sensible to light sources since the real sensor has very bad performance outside were the amount of uv light coming from the sun is sensed as the vcsel emitter

- Sensible to surface colors

- Higlight performance in simulation by using a grid instead of cones to detect distances

Reference

Sources:

[1] *Bitcraze AB Datasheet Multi-ranger deck - Rev 1*

[2] *STMicroelectronics, VL53L1X Datasheet: A new generation, long distance ranging Time-of-Flight sensor based on ST's FlightSense™ technology*

[3] *AN5191, Application note, Using the programmable region of interest (ROI) with the VL53L1X*

[4] *AN5231, Application note, Cover window guidelines for the VL53L1X long-distance ranging Time-of-Flight sensor, Section 5.3*

[5] *Question about mapping using the VL53L1X sensor - Projects / General Guidance - Arduino Forum*